CYBERSECURITY INTERNSHIP

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Role: Cybersecurity Intern | Company: Silverlake Axis Ltd

Duration: 13 July 2026 – 23 October 2026 (15 weeks)

Multimedia University

Internship at a Glance

Fifteen weeks of hands-on security work — measurable outcomes, not just completed tasks

11
security incidents investigated

3,000+
endpoints monitored (SentinelOne EDR)

35+
risks tracked in the company risk register

9+
compliance frameworks mapped & assessed

18
incident types covered in Incident Response playbook

7/7
core tasks delivered and presented to CISO

Company Background & Role Overview

Silverlake Axis Ltd

Item	Detail
Industry	Enterprise technology & software solutions
Focus	Core banking, payment and fintech solutions
Global Presence	Malaysia, Singapore, Thailand and beyond
Scale	~2,000+ employees globally
Security Environment	3,000+ endpoints protected by SentinelOne EDR

Focus: Endpoint Security • Cloud Security • Governance, Risk & Compliance(GRC)

Role: monitored and triaged alerts, investigated incidents, built the risk register, published the Cybersecurity Portal, and collaborated with CISO and IT.

Internship Objectives — All Achieved

#	Objective	Status
1	Gain hands-on experience in cybersecurity operations	✓ Achieved
2	Understand GRC frameworks and risk management	✓ Achieved
3	Develop incident investigation and threat hunting skills	✓ Achieved
4	Learn endpoint security (SentinelOne) at analyst level	✓ Achieved
5	Gain exposure to cloud security (Azure, Huawei)	✓ Achieved
6	Build professional communication and reporting skills	✓ Achieved

Core Tasks Completed — 7/7 Delivered

Task	Key Deliverable
1 – Asset Inventory Review	Reviewed 3,000+ endpoints; identified ~170 duplicate hostnames and ~240 stale devices; escalated for cleanup
2 – Vulnerability Management	Recommended a dedicated vulnerability scanner
3 – Endpoint Security Compliance	Identified 30 unprotected endpoints
4 – Phishing Email Analysis	Analysed a domain-impersonation phishing email; identified SPF/DKIM/DMARC authentication failures
5 – Security Awareness Assessment	Identified top 3 awareness gaps: phishing, AI usage, and incident reporting
6 – Compliance Framework Mapping	Mapped controls to ISO 27001, SOC 1/2/3, PCI DSS, CSA 2024, RMiT, GTRM, PDPA, GDPR, TPRM, BCM, DR, HA
7 – Incident Response Playbook	Collaborated on developing the Incident Response Playbook covering 18 incident types — step-by-step guidance for the IT team

Outcome: all 7 tasks completed with documented deliverables and presented to the CISO.

GRC & Governance Work

Activity	Detail
Risk Register Development	Built and maintained company risk register: 35+ risks with owners, scores and mitigations
Policy Review	Assisted in structured gap analysis of existing policies against target frameworks
External Surface Validation	Validated company domains and public-facing subdomains
USB Whitelist Review	Reviewed USB whitelist entries; recommended revocation of stale/unjustified entries
Website Security Scan	Ran external scan; identified missing security headers

Outcome: audit-readiness foundation built — gaps documented and remediation paths defined.

Incident Investigations — 11 Documented Cases

Alert / Indicator	Classification	Outcome
Dell.TechHub.exe	False Positive	Legitimate Dell update utility — documented
hostguard	False Positive	Legitimate Huawei HSS agent — documented
WinPmem pattern	Threat Hunting	Legitimate bank-initiated memory scan over VPN — pattern closed
PROFITV7 Trojan	Malware (True Positive)	Outbreak across 5 users in one business unit — escalated
kvckiller.sys	Kernel Malware (TP)	Malicious driver from removable media — escalated to CISO; root cause found
ChatGPT.exe	Wiper Malware (TP)	Blocked and mitigated by SentinelOne
FixMe.bat	RDP Persistence (TP)	Blocked and mitigated by SentinelOne
cmd.exe pattern	Pattern Review	Cluster of alerts within 30 minutes — escalated for review
Dell SupportAssist driver	False Positive	Legitimate Dell remediation backup
wailib-sdk	False Positive	Recommended exclusion to reduce alert noise
1321162104.EXE	Credential Dumper	Unsigned exe, encoded PowerShell, log tampering (T1003/T1055/T1105) — escalated; credential-theft response flagged

Outcome: true positives escalated with context, false positives documented, credential-theft response initiated.

Case Study 1 — Malicious Kernel Driver (kvckiller.sys)

From a single EDR alert to a company-wide policy recommendation

1. ALERT

SentinelOne flagged an unsigned kernel driver loading from removable media on an endpoint.

2. INVESTIGATION

Validated the alert as a true positive; traced the driver to a CD-ROM source; checked prevalence across the fleet.

3. ROOT CAUSE

Removable-media controls covered mass-storage device classes only — other device classes bypassed the policy entirely.

4. IMPACT

Kernel-level malware could achieve persistent, high-privilege execution — highest severity class of endpoint threat.

5. RECOMMENDATION

Escalated to CISO with evidence; recommended expanding device control to all removable-media classes and closing the gap.

6. OUTCOME

Root cause accepted; policy enhancement proposed and tracked in the risk register.

Case Study 2 — Credential Dumping Malware

Incident 11: from a High-severity EDR alert to an identity-level response decision

ATTACK CHAIN — WHAT THE MALWARE DID

INITIAL ACCESS
Unsigned executable (1321162104.EXE) launched from user downloads by explorer.exe

EXECUTION
Encoded PowerShell + living-off-the-land binary (LoLBin) proxy execution

DEFENSE EVASION
Event-log tampering, obfuscated code, environmental keying

CREDENTIAL ACCESS
OS credential dumping, API hooking, GUI input capture

MY ACTIONS — MY CONTRIBUTION

1 · ALERT REVIEW
Reviewed MDR case and alert telemetry: unsigned exe on endpoint N7320, High severity, confirmed True Positive. Mapped to MITRE T1003, T1055, T1027, T1685, T1105.

2 · ESCALATED WITH CONTEXT
Escalated to CISO and team with a structured incident summary and recommended response actions — not just a forward.

3 · CREDENTIAL-THEFT CALL
Spotted credential dumping + input-capture indicators and pushed the key question: is a password reset required? (After credential theft: non-negotiable.)

4 · FLAGGED CONTAINMENT GAP
Caught the quarantine failure — only 2/3 files contained — and drove endpoint isolation pending user validation.

OUTCOME

CONTAINMENT
Process killed • 94/94 threats remediated • 2/3 files quarantined • endpoint isolated pending validation

VALIDATION
Mentor confirmed: password reset essential where user interaction occurred — reset initiated

LESSON LOGGED
Mitigation ≠ closure — credential theft demands an identity-level response, not just endpoint cleanup

Cybersecurity Portal — Built & Published

Item	Detail
What	Company-wide Cybersecurity Portal designed, built and published on SharePoint
Sections	Awareness • Best Practices • Policies • Documents • Report an Incident
Audience	All staff, the security team and board members
Status	✓ Live on company SharePoint

Impact: a permanent, central security resource that outlives the internship — company-wide visibility.

SILVERLAKEAXIS Cybersecurity One-Stop Security Hub

Public group

+ New | Page details | Preview | Analytics

CYBERSECURITY PORTAL

STAY SAFE, STAY VIGILANT

Welcome to the SILVERLAKE AXIS Cybersecurity Portal

Security isn't just IT's job it's yours. Every click, every email, every login matters. This portal gives you the tools to stay safe, the context to know why it matters, and the confidence to report when something feels wrong. Because one wrong click can cost millions and 10 seconds of caution can prevent it.







Know The Threats

Security Tips

Cybersecurity Policies & SOPs

CYBER SECURITY BULLETIN

Security Bulletins

Report a Security Incident

Suspect a Security Incident? Report It Immediately to IT Helpdesk: pcsupport@silverlakeaxis.com

Prompt reporting helps minimize the impact of cybersecurity incidents and enables the organisation to respond quickly and effectively. If you notice any suspicious activity, do not ignore it, report it as soon as possible.

What Should Be Reported?

Report incidents such as:

- Suspicious or phishing emails
- Suspected account compromise or unauthorised access
- Malware or ransomware infections
- Lost or stolen company devices
- Accidental exposure or sharing of sensitive information
- Any unusual or suspicious system activity

What Should You Do?

1. **Do not interact further** with the suspicious email, file, or website.
2. **Disconnect your device from the network** if you believe it has been compromised (if instructed by IT or company procedures).
3. **Report the incident immediately** using the approved reporting channel.
4. **Provide relevant details**, such as what happened, when it occurred, and any screenshots if available.

Cybersecurity Awareness

Learn how to recognize cybersecurity threats, protect company information, and follow secure practices in your daily work.



Why Cybersecurity Matters?

Cybersecurity is everyone's responsibility. Every employee plays an important role in protecting company information, systems, and customers from cyber threats. Simple actions such as identifying suspicious emails, protecting credentials, and handling information securely can prevent security incidents.

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Common Cyber Threats

Phishing

Phishing attacks use fraudulent emails, messages, or websites to trick users into revealing information or performing unsafe actions.

Stay protected:

- Check sender details carefully
- Avoid clicking suspicious links
- Verify unexpected requests
- Report suspicious emails



Social Engineering



Security Tips

Simple, everyday actions that protect our company, our clients, and you from strong passwords to spotting phishing emails. Security starts with you.



Keep your device clean. Keep your data safe.

Protect Yourself from Malware



Your shield against cyber risks.

10 CYBER SAFETY MEASURES

Cloud Security Exposure

Activity	Detail
Microsoft Learn – Part 1	Completed 'Describe Cloud Concepts' learning path (3 modules)
Azure Alert Triage	Shadowed cloud security mentor triaging alerts in Microsoft Defender
Huawei Cloud Dashboard	Learned Huawei Cloud console, alerting and dashboard views
MITRE + OWASP Mapping	Learned mapping cloud alerts to MITRE ATT&CK and OWASP frameworks
Multi-Cloud Exposure	Gained working awareness across Azure and Huawei Cloud environments
Cloud Security Workflow	Learned how cloud alerts are verified, escalated and remediated end-to-end

Key learning: cloud alert triage follows the same validation mindset as endpoint — confirm, assess impact, escalate.

Endpoint Security & SentinelOne Monitoring

Activity	Detail
Alert Triage	Reviewed and triaged EDR alerts daily; separated noise from genuine threats
Escalation	Escalated true positives to CISO with evidence, context and recommendations
Root-Cause Analysis	Identified the removable-media control gap behind the kernel-driver incident
Pattern Recognition	Spotted WinPmem, PROFITV7 and kvckiller patterns before they became noise

Key achievement: Conducted deep-dive EDR alert triage, independently validated true vs. false positives, and escalated with technical context.

Skills Gained

Technical Skills	GRC Skills	Professional Skills
SentinelOne EDR alert triage & investigation	Risk register management	Professional communication
SharePoint	ISO 27001 / SOC 2 mapping	MDR vendor coordination
Threat hunting & pattern analysis	Policy review & gap analysis	Report writing
Incident response & root-cause analysis	Statement of Applicability development	Presentation skills
Cloud security fundamentals (Azure, Huawei)	Framework & control mapping	User outreach

Tools: SentinelOne • Microsoft Defender • Azure Portal • Huawei Cloud • Pentest-Tools • VirusTotal • Excel

Key Achievements

Achievement	Impact
Investigated 11 documented security incidents	Real threats handled — malware, kernel drivers, phishing, persistence
Built & published the Cybersecurity Portal	Company-wide resource, live on SharePoint
Built risk register with 35+ risks	Governance tool adopted by the security team
Identified removable-media control gap	Root cause of the kernel-driver incident — fix recommended
Prepared CISO-facing threat reports	Monthly SentinelOne reporting, stale-device and USB whitelist reviews
Earned multi-cloud exposure (Azure + Huawei)	Foundation for cloud security career path
Delivered 7/7 core tasks	Foundation for GRC
Contributed to the 18-type Incident Response Playbook	Step-by-step response guidance for IT team; institutional knowledge that outlives the internship

One-liner: delivered practical, measurable outcomes — not just completed tasks.

Problems Encountered & How They Were Addressed

Problem	Response
Removable-media policy coverage gaps	Identified device-class bypass; recommended expanding controls to all removable media
Excessive false positives (wailib-sdk)	Recommended tuned exclusion to reduce noise and alert fatigue
~240 stale devices in inventory	Escalated to CISO — reduces licence waste and incident confusion
MDR quarantine failure on read-only media	Documented root cause; recommended compensating controls

Improvements proposed: full device-class coverage • BEC playbook + detection rule • quarterly exclusion reviews • quarterly asset cleanup process.

Strengths & Growth Areas

Strengths	Growth Area → Plan
Analytical thinking — 11 incidents investigated; patterns and root causes identified	Deep Visibility (S1QL) queries → continued practice & hunting
Initiative — built portal, risk register and escalated incidents beyond assigned tasks	Hands-on cloud depth → labs
Communication — CISO briefings, MDR team, user outreach	Offensive security → CPTS after internship
Adaptability — moved across GRC, endpoint security and cloud workstreams	Real-world pentest experience → home lab + CTF competitions

Traits: analytical • self-starting • communicative • adaptable • consistent across 15 weeks.

Applying University Knowledge

Subject / Knowledge	How It Was Applied
Cybersecurity Fundamentals	Alert triage, incident investigation and threat hunting
Operating Systems (Windows/Linux)	Malware, kernel-driver and endpoint alert investigation
Research & Documentation	Produced CISO reports, internship logs and this portfolio

Key takeaway: university provided the theory — the internship turned it into practical capability.

Conclusion

Area	Summary
Tasks	All 7 core tasks delivered with documented outcomes
Incidents	11 documented cases — malware, kernel drivers, phishing, credential dumping, persistence
GRC	Risk register built, frameworks mapped, policies reviewed
Portal	Company-wide Cybersecurity Portal live on SharePoint
Endpoint Security	Monitored EDR alerts, validated threats, and managed escalations
Cloud	Multi-cloud awareness across Azure and Huawei

“I didn't just complete tasks — I investigated real incidents, built governance tools and improved the company's security posture. I leave with proof I can operate, not just learn.”

Lessons Learned

Lesson	Why It Matters
Containment is not closure	Killing a process isn't enough — root cause must be fixed to prevent recurrence
Not every alert is a threat	False positives are normal — but their patterns still teach you
Context changes everything	WinPmem looked malicious until we learned it was bank-initiated scanning
Policy gaps are real risks	A partial control is often worse than no control — it creates false confidence
Cloud is the future	Multi-cloud exposure is essential for modern security careers
Communication is key	Soft skills carry as much weight as technical skills
Documentation proves work	Records, reports and a portfolio build your reputation

Key takeaway: security is not just tools — it's thinking, context and communication.

THANK YOU

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